ConvPE: Convolutional Position Embedding Improves Transformers

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ABSTRACT

ConvPE (Convolutional Position Embedding) substantially improves both the performance and efficiency of Transformer models. At the cost of only 2.64% more parameters and a 2.70% increase in forward-pass time, it achieves a 2.82% reduction in language-modeling loss and substantial improvements on most evaluation benchmarks. It also requires only 68.7% training tokens to reach the baseline's final training loss. ConvPE injects position information through windowed attention-score biases generated by a lightweight convolution over token representations, making it compatible with various Transformer architectures and naturally extendable to high-dimensional modalities. The code is publicly available at `https://github.com/wzhagi/convpe`.

Keywords Position Embedding · Language Models · Transformers · Attention · Vision Transformers

1 Introduction

Transformers[1] are insensitive to input order. However, word order is crucial for language understanding. Explicit or implicit position embeddings are therefore necessary to inject order information into the model[2,3,4].

This paper proposes ConvPE, a novel and efficient position embedding scheme. ConvPE embeds position information into attention score biases, trading a minor parameter overhead for significant performance gains, as shown in Figure 1. Its design allows the model to switch freely between different attention masks and is compatible with various Transformer architectures. It also naturally supports high-dimensional modalities with n-dimensional convolution.

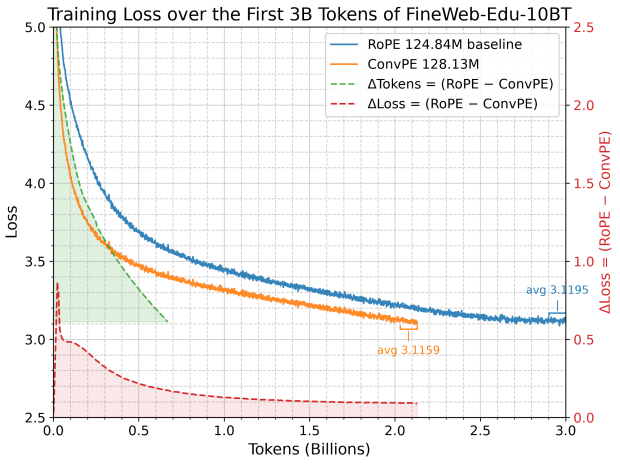


Figure 1

The main contributions of this paper are as follows:

1. We propose the ConvPE architecture that combines high performance and broad compatibility.

2. We study the influence of ConvPE hyperparameters and architecture.

3. We verify the effectiveness of ConvPE in language model pretraining.

4. We verify the effectiveness of ConvPE2D in visual model training.

5. We developed high performance TileLang[5] kernels for ConvPE.

2 Introduction

Transformers[1] are insensitive to input order, whereas word order is crucial for language understanding. Explicit or implicit position embeddings are therefore necessary to inject order information into the model[2,3,4].

This paper proposes ConvPE, a simple and efficient position-embedding scheme that generates windowed attention-score biases with a lightweight convolution. ConvPE can switch freely between different attention masks, be combined with RoPE, support KV-cache concatenation, and naturally extend to high-dimensional modalities with n-dimensional convolution.

We evaluate ConvPE in language-model pretraining and vision transformers. ConvPE outperforms the RoPE baseline across a range of hyperparameter configurations. In pretraining on the first 3B tokens of FineWeb-Edu-10BT, ConvPE adds only 2.64% parameters and 2.70% forward-pass time, while reducing the final training loss from 3.1195 to 3.0316, a 2.82% reduction and improving most downstream benchmarks. More importantly, ConvPE reaches the RoPE baseline's final loss with 2.06B rather than 3B tokens. Despite a roughly 7% per-step wall-clock overhead, this translates to an estimated 26.5% net reduction in wall-clock training time to match the baseline loss. ConvPE remains language-modeling loss comparatively stable under context extension up to 4K, and outperforms the RoPE baseline on several RULER tasks. Finally, ConvPE2D improves ViT-Tiny[6] accuracy on CIFAR-10 and CIFAR-100[7].

The main contributions of this paper are as follows:

- We propose ConvPE, a convolutional attention-score-bias position embedding that combines strong performance with broad compatibility.
- We study the influence of ConvPE hyperparameters and architecture choices, and identify a simple default configuration with a favorable performance–efficiency balance.
- We verify the effectiveness of ConvPE in language-model pretraining, including token efficiency, downstream evaluations, and context-extension behavior.
- We verify the effectiveness of ConvPE2D in vision transformers on CIFAR-10 and CIFAR-100.
- We develop high-performance TileLang[5] kernels for ConvPE.

3 Background

3.1 Attention Mechanism

Given the input query $Q \in \mathbb{R}^{S_q \times D_k}$, key $K \in \mathbb{R}^{S_{kv} \times D_k}$, and value $V \in \mathbb{R}^{S_{kv} \times D_v}$, the computation of scaled dot-product attention[1] $O \in \mathbb{R}^{T \times D_v}$ is:

$$O = \mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \mathbf{B}\right) V$$

T is the query sequence length. S is the key/value sequence length. $\mathbf{B} \in \mathbb{R}^{S_q \times S_{kv}}$ is the attention score bias. Causal models introduce a causal mask $M \in \mathbb{R}^{S_q \times S_{kv}}$ that sets the attention score bias $\mathbf{B}$ for invisible positions to $-\infty$. With batch size and multi-head settings, Q, K, V, O additionally have B, H dimensions.

3.2 Position Embedding

Existing position embedding schemes can be roughly divided into five categories:

1. **Absolute Position Embedding**[1]: adds a fixed positional bias, usually sine waves, to the input vector. It is simple to implement, but its performance is usually limited.

2. **Learnable Positional Bias**[2,3]: adds a learnable positional bias to the input sequence. It is typically adopted in limited-window settings, such as spatial dimensions in vision models or fixed-rate compression.

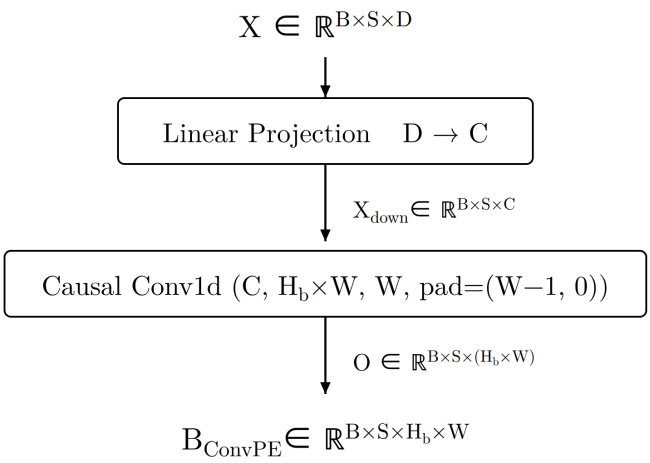
3. **Rotary Position Embedding**[4]: encodes position as a rotation matrix and applies it to queries and keys, so that attention scores depend on relative positions. RoPE has been widely adopted in modern large language models, enabling long-context extension via frequency scaling, such as YaRN[8].

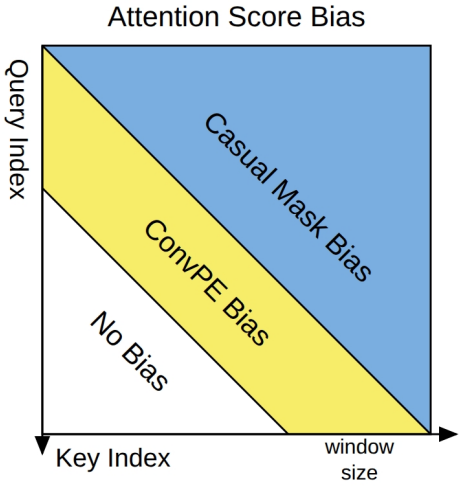
4. **Attention Score Bias**: adds a positional bias B to attention scores[9 10 11]. RoPE can be viewed as a rotation-based equivalent, as it also makes attention scores depend on relative positions.

5. **No Position Embedding (NoPE)**[12]: adds no explicit position embeddings; instead, it relies on model structure and causal attention masks to implicitly learn positional information requiring no explicit Position Embedding[13].

4 Method

Given token representations $X \in \mathbb{R}^{B\times S\times D}$, ConvPE first projects them to a small channel dimension $X_{down} \in \mathbb{R}^{B\times S\times C}$, then use a convolution to generates windowed attention score bias $\mathbf{B}_{ConvPE} \in \mathbb{R}^{B\times S\times H_b\times W}$ for self-attention keys within the convolution window.





$$\mathbf{B}^{b,h,s_q,s_{kv}} = \begin{cases} \mathbf{B}^{b,s_q,h\%H_b,w}_{\text{ConvPE}}, & w \in [0, W-1] \text{ where } w = s_{kv} - s_q + W - 1 \\ 0, & \text{otherwise.} \end{cases}$$

The hyperparameters are:

- W: convolution window size.
- C: out features of linear projection and in channels of convolution.
- H_b: number of score bias heads.

The convolution takes C input channels, uses a kernel of size W, and outputs $H_b \times W$ channels. The projection has $D \times C$ parameters and the convolution has $W^2 \times C \times H_b$ parameters; the computation (MACs) per token is the same as the parameter count.

To reduce the convolution parameter count, we introduce three additional hyperparameters:

- Groups: number of convolution groups;
- Share: whether to share weights across convolution groups;
- Rank: rank used in the low-rank convolution.

5 Experiments

All models in our experiments use Multi-Query Attention (MQA)[14], with $K = V$[15]. Each attention layer has $D_{model} \times D_{head} \times (2 \times N_{heads} + 1)$ parameters. Each feed-forward layer uses SwiGLU[16] and has $8 \times D^2_{model}$ parameters.

5.1 Hyperparameter Search

We train each model for one epoch on WikiText-103[17] and record the validation loss. Training uses the Hugging Face Trainer[18], AdamW[19], and the GPT-2 tokenizer[3]. The training hyperparameters are: batch size 4, sequence length 1024, learning rate 3×10^{-4} with cosine learning-rate scheduler, and seed 42.

We train two types of backbone models, a tiny backbone with $N_{layers} = 12$, $D_{model} = 192$, $N_{heads} = 3$, $D_{head} = 64$, and a base backbone with $N_{layers} = 12$, $D_{model} = 768$, $N_{heads} = 12$, $D_{head} = 128$. The model name format is "`backbone RoPE/NoPE` W C Rank H_b Groups Share". For example, "`tiny rope w128d6r128h3g1s`" denotes a tiny backbone using RoPE with $W = 128$, $C = 6$, Rank $= 128$, $H_b = 3$, Groups $= 1$, and Share weights. For simplicity, model names may omit the default configuration: `NoPE`, `r0`(no low-rank convolution), `h1` ($H_b = 1$), `g1`(Groups $= 1$), and `s0` (no weight sharing).

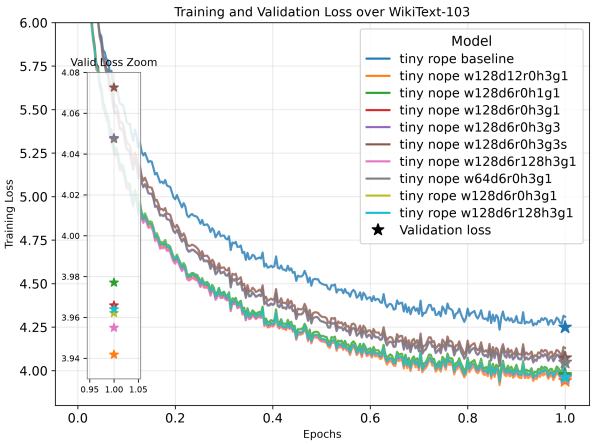


Figure 2: Tiny model training curve on WikiText-103

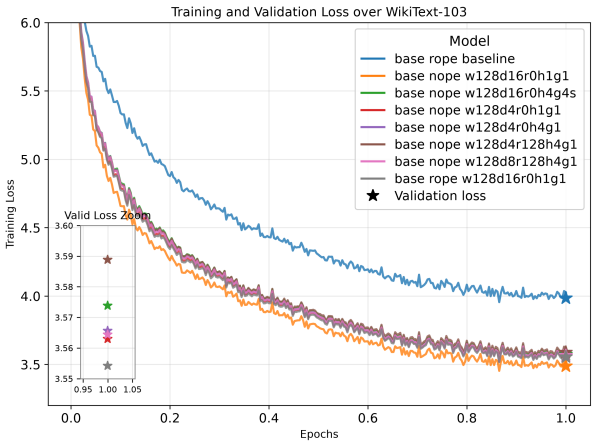


Figure 3: Base model training curve on WikiText-103

First, we train tiny models with hyperparameters around "`nope w128d6r0h3g1`". Figure 2 shows that all ConvPE variants outperform the RoPE baseline. The largest gains come from increasing the window size and the projection dimension. Using grouped convolution causes the largest performance drop. RoPE integration and other parameter-saving methods show minor differences from "`nope w128d6r0h3g1`".

We then train base model with hyperparameters around "`nope w128d4r0h4g1`" and "`nope w128d16r0h1g1`". Figure 3 shows that $H_b = 1$ without any parameter-saving technique achieves the best balance between performance and parameter efficiency.

In additional experiments, we also try using X , Q[11] and KV as the convolution input, but performance drops.

5.2 Pretraining

We choose "`base w128d16`" as the ConvPE variant configuration for performance and throughput considerations. We train the RoPE baseline `base RoPE` and the ConvPE variant from scratch on the first 3B tokens of FineWeb-Edu-10BT. We use Muon[20] to optimize the 2D parameters, excluding the embedding layer and the language head, with a learning rate of 3×10^{-3}, while Adam[21] optimizes the remaining parameters at a learning rate of 3×10^{-4}, both using a cosine learning-rate scheduler. The training hyperparameters are: batch size 4, gradient accumulation steps 4, sequence length 1024, and seed 42.

Figure 4 shows that ConvPE converges faster and achieves a lower final loss compared to the RoPE baseline. We implement high-performance Flash MQA kernels in TileLang and benchmark two configurations: `base RoPE` with 124.84M parameters and `base w128d16` with 128.13M parameters (+2.64%), using a batch size of 4 and a sequence length of 1024. While the forward-pass time increases by 2.70%, slightly exceeding the parameter increase, and the overall wall-clock time per step increases by approximately 7%, the final training loss drops from 3.1195 to 3.0316 (-2.82%). According to scaling laws[22,23], such a substantial loss reduction would typically require a model with $\sim 50\%$ more parameters. More importantly, despite the 7% per-step overhead, the number of training tokens required to match the RoPE baseline's final loss decreases from 3B to 2.06B, a 31.3% reduction. This translates to a net reduction in total wall-clock training time of approximately 26.5%.

5.3 Evaluations

This section evaluates pretrained ConvPE and RoPE baseline on pretraining and context-extension benchmarks without any fine-tuning.

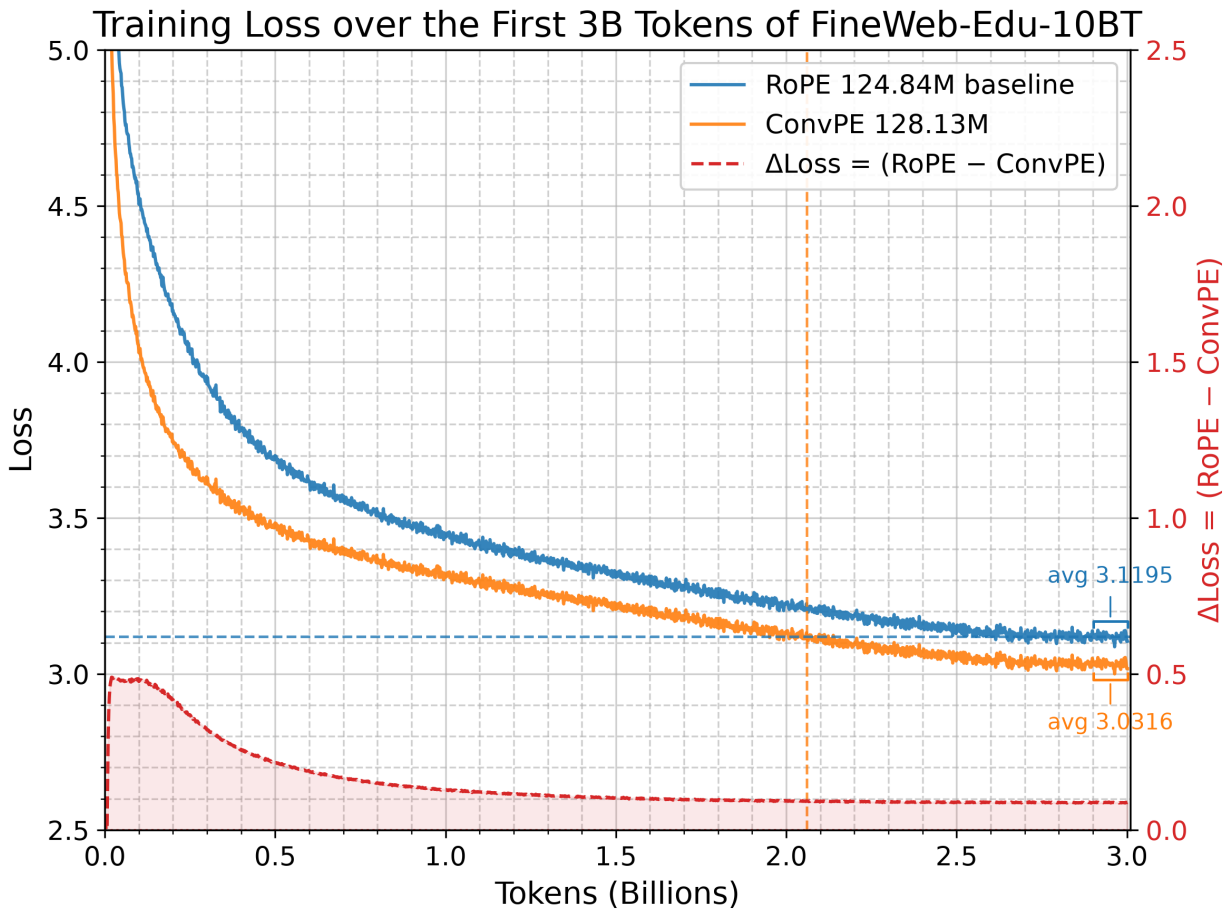


Figure 4: Pretraining curve on first 3B tokens of FineWeb-Edu-10BT

5.3.1 Pretraining Evaluations

Dataset	Metric	RoPE	ConvPE	Random
wikitext	perplexity	46.56	**42.05**	-
lambada	perplexity, acc	118.10, 0.2507	**88.49**, **0.2604**	-
hellaswag	acc, acc norm	0.2894, 0.3148	**0.2949**, **0.3272**	0.25
xstorycloze_en	acc	0.5460	**0.5526**	0.5
piqa	acc, acc norm	0.6251, 0.6208	**0.6338**, **0.6235**	0.5
arc_easy	acc, acc norm	0.5114, 0.4487	**0.5421**, **0.4701**	0.25
winogrande	acc	0.5091	**0.5193**	0.5
boolq	acc	0.5676	**0.5893**	0.5
truthfulqa_mc1	acc	0.2399	0.2375	**0.25**

Table 1: Evaluation results of pretraining models

Table1 shows that ConvPE consistently improves language-modeling performance over the RoPE baseline. On WikiText, perplexity drops from 46.56 to 42.05, and on LAMBADA perplexity decreases substantially from 118.10 to 88.49, with accuracy also improving from 0.2507 to 0.2604. ConvPE also outperforms RoPE on most downstream benchmarks. The gains are consistent across metrics. The main exception is TruthfulQA-MC1, where both models fall below the random-choice baseline, indicating that this benchmark remains difficult for both models. Overall, the pretraining results support ConvPE as a stronger positional encoding than RoPE for the evaluated language tasks.

Context	RoPE	ConvPE
1K	3.1247	**3.0312**
2K	3.1684	**3.0021**
4K	4.2690	**3.0278**
8K	5.6443	**3.3575**

Table 2: Validation Loss on 1M Training Tokens

model	params	CIFAR-10	CIFAR-100
RoPE baseline	5.346M	0.910	0.627
ConvPE2D `w3d4`	5.359M	0.931	**0.695**
ConvPE2D `w3d8`	5.372M	**0.933**	0.683
ConvPE2D `w5d4`	5.402M	0.929	0.682
ConvNet	4.335M	**0.945**	**0.737**

Table 3: Validation accuracy on CIFAR-10 and CIFAR-100

Task	Context	RoPE	ConvPE
niah_single_1	1K	0.614	**0.890**
	2K	0.004	**0.722**
	4K	0.000	**0.314**
niah_single_2	1K	0.612	**0.664**
	2K	0.000	**0.038**
niah_single_3	1K	0.090	**0.180**
	2K	0.000	**0.014**
niah_multikey_1	1K	0.198	**0.200**
	2K	0.118	**0.194**
	4K	0.000	0.002
niah_multikey_2	1K	0.002	0.000
niah_multikey_3	1K	0.000	0.008
niah_multiquery	1K	0.104	**0.186**
	2K	0.029	**0.179**
	4K	0.000	0.000
niah_multivalue	1K	0.149	**0.187**
	2K	0.127	**0.186**
	4K	0.000	0.000
ruler_vt	1K	**0.032**	0.008
	2K	0.000	**0.052**
	4K	0.000	**0.107**
ruler_cwe	1K	0.002	0.002
ruler_fwe	1K	0.001	**0.012**
ruler_qa_squad	1K	0.148	**0.191**
	2K	0.078	**0.091**
	4K	0.017	**0.052**

Table 4: Ruler Evaluations

5.3.2 Context Extension Evaluations

We first evaluate language-modeling performance using 1M tokens from the training dataset with increasing context length. Both models are pretrained on 1K-token contexts. Table 2 shows a clear advantage for ConvPE in language modeling as context length increases. RoPE degrades sharply beyond 2K, whereas ConvPE remains comparatively stable up to 4K and degrades much more gracefully at 8K. This indicates that ConvPE provides better context-length extrapolation and preserves model performance under longer contexts.

We then evaluate both models on RULER tasks. Table 4 shows that ConvPE generally improves over RoPE on several long-context retrieval and reasoning tasks, particularly at 1K and 2K contexts. ConvPE even performs well at 4K contexts on some tasks. Some tasks remain near zero for both models, indicating that these tasks are too difficult for 125M-parameter models. Although ConvPE language-modeling loss remains stable up to 4K, the downstream tasks show that zero-shot context extension is still far from solved.

5.4 2D Position Embedding

To produce 2D positional embeddings, we use 2D convolutional layers that take a 2D window input and produce a 2D window output. We use ViT-Tiny[6] models with Multi-Head Attention (MHA) and the configuration $N_{\text{layers}} = 12$, $D_{\text{model}} = 192$, $N_{\text{heads}} = 3$, and $D_{\text{head}} = 64$. We train the models on CIFAR-10 and CIFAR-100[7] for 100 epochs with patch shape 4×4.

Table 3 shows that ConvPE transfers effectively to vision transformers and outperforms the RoPE baseline. The influence of hyperparameters requires further study on larger models and datasets.

6 Conclusions and Future Work

ConvPE delivers substantial performance gains at a minor parameter and computational cost, providing a powerful alternative for positional embeddings. We adopt $H_b = 1$ without any parameter-saving technique as the default

configuration, which achieves the best balance among performance, parameter efficiency, and computational cost. ConvPE is compatible with RoPE, various attention masks, and KV-cache concatenation, and it naturally supports multidimensional extensions.

Future work can explore the following directions:

- **Kernel Optimization.** Further optimize the backward kernel and develop additional kernels for a wider range of hardware.
- **Scaling behavior.** Investigate ConvPE with larger Transformers and datasets, and identify configuration patterns at scale.
- **Long-context modeling.** Evaluate ConvPE in long-context settings, where positional generalization is particularly important.
- **Architectural integration.** Combine ConvPE with Sliding Window Attention (SWA)[24], Compressed Sparse Attention (CSA)[15], and other Transformer variants.
- **Multidimensional modalities.** Extend ConvPE to high-dimensional modalities such as vision, speech, and multimodal data.

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